WORKED SOLUTION

STEP II 2023, Question 5

Pure • Recurrence Relations and Convergence

STEP II 2023, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.7, *Six Digits or Six Hundred*: https://beyondstep.keithmcnulty.org/ch02/newton-speed.html

The key idea

Both recurrences are machines for approximating $\sqrt{2}$, and the question is about how fast the error shrinks. The trick in each part is to measure the error in the right way: in (i) look at $x_n^2 - 2$, which gets multiplied by a factor of size at most $\frac{1}{4}$ (and changes sign) at every step; in (ii) look at $y_n - \sqrt{2}$, which gets *squared* (roughly) at every step. Multiplying by a constant gives steady progress; squaring doubles the number of correct digits each time, and that is why part (ii) reaches 10^{-600} in the same ten steps.

Solution

Part (i)

(a) We can rewrite the recurrence as

$$x_{n+1} = \frac{(x_n + 1) + 1}{x_n + 1} = 1 + \frac{1}{x_n + 1}.$$

We have $x_0 = 1 \geq 1$. If $x_n \geq 1$, then $x_n + 1 > 0$, so $\frac{1}{x_n+1} > 0$ and $x_{n+1} > 1$. By induction, $x_n \geq 1$ for all n.

(b) Put everything over a common denominator:

$$\begin{aligned} x_{n+1}^2 - 2 &= \frac{(x_n+2)^2 - 2(x_n+1)^2}{(x_n+1)^2} \\ &= \frac{x_n^2 + 4x_n + 4 - 2x_n^2 - 4x_n - 2}{(x_n+1)^2} \\ &= \frac{-(x_n^2-2)}{(x_n+1)^2}. \end{aligned}$$

The denominator is positive, so $x_{n+1}^2 - 2$ and $x_n^2 - 2$ have opposite signs. (Neither is ever zero: every x_n is rational, by induction, and $\sqrt{2}$ is irrational.)

Taking moduli, and using $x_n + 1 \geq 2$ from (a), so that $(x_n+1)^2 \geq 4$,

$$|x_{n+1}^2 - 2| = \frac{|x_n^2 - 2|}{(x_n+1)^2} \leq \tfrac{1}{4}|x_n^2 - 2|,$$

as required.

(c) Since $x_0^2 - 2 = -1$, applying (b) ten times gives

$$|x_{10}^2 - 2| \leq \left(\tfrac{1}{4}\right)^{10} |x_0^2 - 2| = \frac{1}{4^{10}} = \frac{1}{1\,048\,576} < 10^{-6}.$$

For the sign: $x_0^2 - 2 < 0$ and the sign flips at every step, so $x_n^2 - 2$ is negative for every even n. In particular $x_{10}^2 - 2 < 0$, and so

$$-10^{-6} < x_{10}^2 - 2 < 0, \qquad \text{that is,} \qquad 2 - 10^{-6} \leq x_{10}^2 \leq 2.$$

Part (ii)

(a) First, every y_n is positive: $y_0 = 1 > 0$, and if $y_n > 0$ then $y_{n+1} = \frac{y_n^2+2}{2y_n} > 0$. So the recurrence never divides by zero. Now

$$y_{n+1} - \sqrt{2} = \frac{y_n^2 + 2 - 2\sqrt{2}\,y_n}{2y_n} = \frac{(y_n - \sqrt{2})^2}{2y_n},$$

since $(y_n - \sqrt{2})^2 = y_n^2 - 2\sqrt{2}\,y_n + 2$. This is the required identity.

For the deduction: $y_0 = 1$. For $n \geq 1$, the identity (applied with $n-1$) shows that $y_n - \sqrt{2}$ is a square divided by the positive number $2y_{n-1}$, so $y_n - \sqrt{2} \geq 0$, and hence $y_n \geq \sqrt{2} > 1$. So $y_n \geq 1$ for all $n \geq 0$.

(b) Write $e_n = y_n - \sqrt{2}$ and $r = \dfrac{\sqrt{2}-1}{2}$. From (a), $e_n \geq 0$ for $n \geq 1$, and since $y_n \geq 1$,

$$e_{n+1} = \frac{e_n^2}{2y_n} \leq \frac{e_n^2}{2}.$$

We prove $e_n \leq 2r^{2^n}$ for $n \geq 1$ by induction.

Base case $n = 1$. $y_1 = \frac{3}{2}$, so $e_1 = \frac{3}{2} - \sqrt{2}$, while

$$2r^2 = 2 \cdot \frac{(\sqrt{2}-1)^2}{4} = \frac{3 - 2\sqrt{2}}{2} = \tfrac{3}{2} - \sqrt{2}.$$

So the bound holds with equality.

Inductive step. Suppose $0 \le e_n \le 2r^{2^n}$ for some $n \ge 1$.

? **Ask yourself** Why are we allowed to square both sides of $e_n \le 2r^{2^n}$?

Squaring preserves an inequality between *non-negative* numbers, and $e_n \ge 0$ is exactly what part (a) gave us. (This is why the bound starts at $n = 1$, not $n = 0$: $e_0 = 1 - \sqrt{2}$ is negative.) So

$$e_{n+1} \le \frac{e_n^2}{2} \le \frac{\left(2r^{2^n}\right)^2}{2} = \frac{4r^{2^{n+1}}}{2} = 2r^{2^{n+1}},$$

which completes the induction.

(c) The lower bound $y_{10} \ge \sqrt{2}$ was shown in (a). For the upper bound, $\sqrt{2} - 1 < \frac{1}{2}$ gives $0 < r < \frac{1}{4}$, so by (b)

$$y_{10} - \sqrt{2} \le 2r^{2^{10}} = 2r^{1024} < 2\left(\tfrac{1}{4}\right)^{1024} = 2 \cdot 2^{-2048} = 2^{-2047}.$$

To compare this with a power of 10, use $2^{10} = 1024 > 10^3$:

$$2^{2047} > 2^{2040} = \left(2^{10}\right)^{204} > \left(10^3\right)^{204} = 10^{612} > 10^{600}.$$

Hence $y_{10} - \sqrt{2} < 2^{-2047} < 10^{-600}$, and altogether

$$\sqrt{2} \le y_{10} \le \sqrt{2} + 10^{-600}.$$

? **Ask yourself** Where does the extraordinary speed in (ii) come from?

In (i) the error is multiplied by at most $\frac{1}{4}$ each time, so ten steps gain only about six decimal places. In (ii) the error is (roughly) squared each time, so the exponent 2^n doubles: ten steps take us from an error of about $\frac{1}{10}$ to one below 10^{-600}.

Checks

- The first few terms: $x_1 = \frac{3}{2}$, $x_2 = \frac{7}{5}$, $x_3 = \frac{17}{12}$, and $x_n^2 - 2$ goes -1, $\frac{1}{4}$, $-\frac{1}{25}$, $\frac{1}{144}$: alternating, and shrinking by at least a factor of 4 each time.
- Computing exactly with fractions, $x_{10}^2 \approx 2 - 3.0 \times 10^{-8}$, comfortably inside the window $[2 - 10^{-6}, 2]$.
- $y_2 = \frac{17}{12}$, so $y_2 - \sqrt{2} \approx 0.00245$, against the bound $2r^4 \approx 0.00368$. A computer check with exact fractions gives $y_{10} - \sqrt{2} \approx 10^{-783}$, well below the guaranteed $2^{-2047} \approx 10^{-616}$.

Common slips

- Forgetting the sign in (i)(c): the modulus bound alone gives $2 - 10^{-6} \leq x_{10}^2 \leq 2 + 10^{-6}$; the upper bound 2 needs the alternating signs.
- Squaring $e_n \leq 2r^{2^n}$ without first knowing $e_n \geq 0$.
- Stopping at 2^{-2047} without actually comparing it to 10^{-600}; the inequality $2^{10} > 10^3$ is the cleanest way.

Beyond STEP

The recurrence in (ii) is Newton's method applied to $f(y) = y^2 - 2$, and the identity in (ii)(a) is the statement that Newton's method converges *quadratically* near a simple root, whereas (i) is an example of *linear* convergence with rate at most $\frac{1}{4}$. The numbers x_n are also the convergents of the continued fraction for $\sqrt{2}$. Orders of convergence and fixed-point iteration are standard topics in a first numerical analysis course, and the analysis behind them is developed carefully in Bartle and Sherbert, *Introduction to Real Analysis*.

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